

Past, Present and (Possible) Futures of Valuations and Risk Management under IBORs evolution

Benchmark Rates Reform: Valuations, Discounting and Forward Risk
London, 15 March 2019

Marco Bianchetti, Intesa Sanpaolo, Financial and Market Risk Management, and University of Bologna
Marco Scaringi, Intesa Sanpaolo, Financial and Market Risk Management

Disclaimer and Acknowledgments

Disclaimer

The views and the opinions expressed in this document are those of the authors and do not represent the opinions of their employers. They are not responsible for any use that may be made of these contents. This document is written for informative purposes only, it is not intended to influence any investment decisions or promote any product or service.

Acknowledgments

We acknowledge fruitful discussions with many colleagues in Intesa Sanpaolo and Banca IMI.

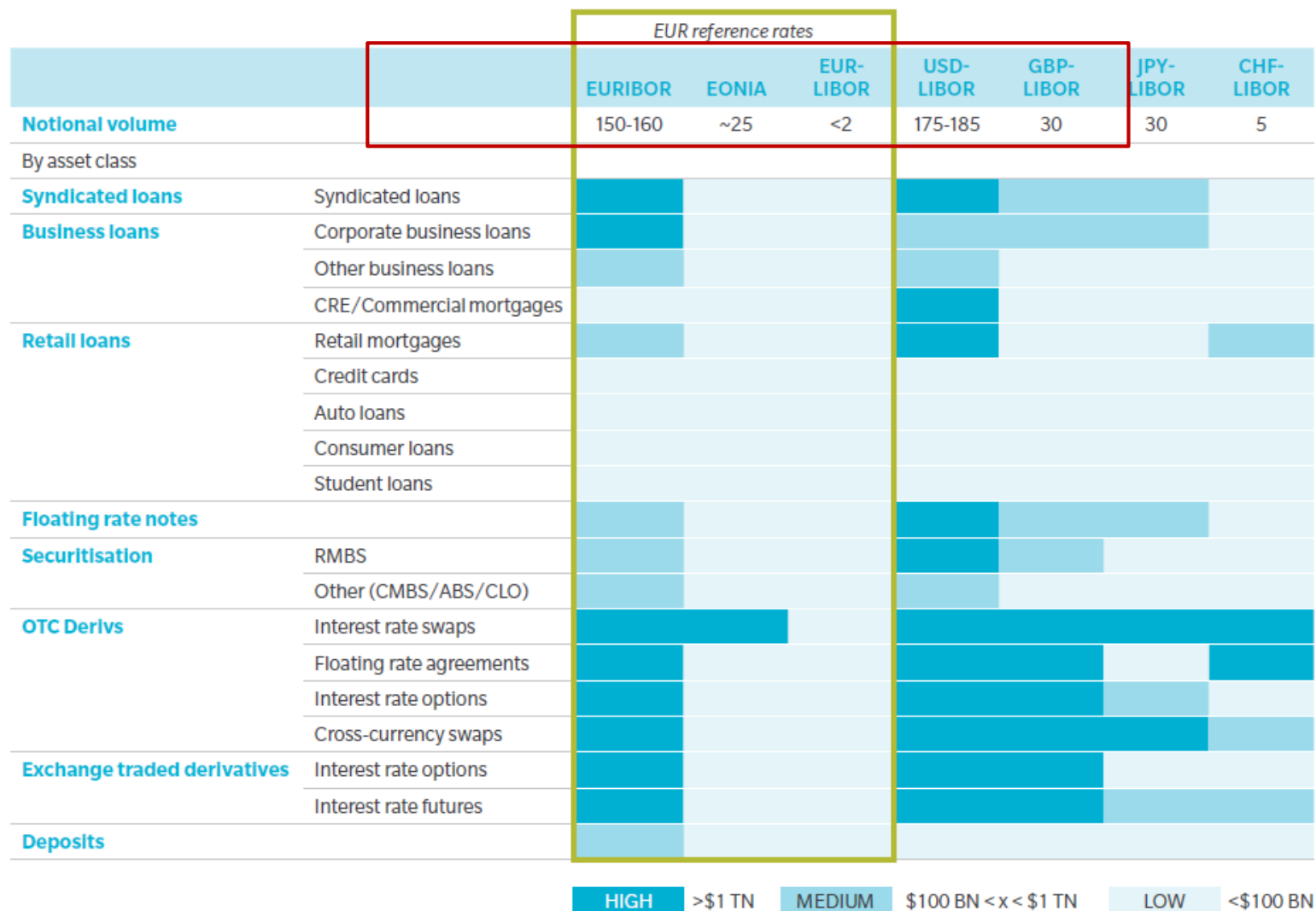
Summary

1. Introduction
2. Classic vs Modern Benchmark Rates
 - SOFR
 - ESTER
 - EURIBOR
 - Transition Impacts
3. Pricing and Risk Management
 - The fundamental role of CCPs
 - Risk Management Issues
 - Pricing impacts
4. Focus on XVAs
5. Bye-Bye multi-curves?
6. References
7. Appendix: funding spread



1: Introduction

Interest rates are everywhere



Source: Oliver Wyman analysis, data as available as of December 2017 and updated to reflect estimates from the 2nd and 3rd meetings of the working group on euro risk-free rates

2: Classic vs Modern Benchmark Rates

Classic interest rates

Jurisdiction					
Rate	Euro OverNight Index Average (EONIA)	Federal Funding Effective Rate (Fed Fund)	Sterling OverNight Index Average (SONIA)	Swiss Average Rate OverNight (SARON)	Tokyo OverNight Average Rate (TONAR)
Rate administrator	European Money Market Institute (EMMI)	Federal Reserve Bank of NY	Wholesale Market Brokers Association (WMBA)	Eurex Zurich Ltd	Bank of Japan
Live	4 January 1999	NA	March 1997	25 August 2009	11 April 1996
Tenor	Overnight	Overnight	Overnight	Overnight	Overnight
Secured	No	No	No	Yes	No
Publication time	T (19:00 CET)	T+1 (9:00 ET)	T (18:00 GMT)	T (12:00, 16:00, 18:00 CET)	T (17:15 JST)
Panel	28 EU banks	Institutions holding reserve balances at the NY Fed	WMBA members	--	--
Market	EUR Interbank lending transactions conducted in the EU and EFTA countries	USD borrowings of reserve balances between depository institutions at NY Fed	GBP borrowings brokered by WMBA members between London counterparties.	CHF interbank repo transactions	JPY uncollateralised overnight call transactions
Data	Data contributed from panel banks before 18:00 CET	Data from at the Domestic Trading Desk of the NY Fed	Data from WMBA members between 00:00 and 15.15 GMT, size > 25 mln GBP	Data from the order book of SIX Repo Ltd electronic trading platform	Data from main money market brokers
Formula	Volume-weighted average	Volume-weighted average	Volume-weighted average	Volume-weighted average	Volume-weighted average

2: Classic vs Modern Benchmark Rates

Modern interest rates

Jurisdiction					
Rate	Euro Short Term Rate (ESTER)	Secured Overnight Financing Rate (SOFR)	Reformed SONIA	SARON	TONAR
Working Group	Working Group on Euro RFR for the Euro area	Alternative Reference Rates Committee	Working Group on Sterling Risk-Free Reference Rates	National Working Group on Swiss Franc Ref. Rate	Study Group on Risk-Free Reference Rates
Rate administrator	European Central Bank	Federal Reserve Bank of NY	Bank of England	SIX Swiss Exchange	Bank of Japan
Live	Pre-ESTER: 15/3/2017 ESTER: Oct. 2019	3 April 2018	23 April 2018	25 August 2009	11 April 1996
Transition plan	No	Yes	No	No	No
Tenor	Overnight	Overnight	Overnight	Overnight	Overnight
Secured	No	Yes	No	Yes	No
Publication time	T+1 (9:00 CET)	T+1 (8:30 ET)	T+1 (9:00 GMT)	T (12:00, 16:00, 18:00 CET)	T (17:15 JST)
Market	EUR borrowing cost in the wholesale sector from MMSR banks	USD borrowing collateralized by Treasury Securities	GBP borrowing wholesale bilateral or broker transactions	CHF interbank repo transactions	JPY uncollateralized overnight call rate market
Data	Data from MMSR, size > 1 mln€	Data from BNYM and DTCC	Data from Sterling Money Market daily collection, between 00:00 and 18:00 GMT size > 25 millions GBP	Data from the order book of SIX Repo Ltd electronic trading platform	Data from main money market brokers
Formula	Volume-weighted trimmed average (25%)	Volume-weighted trimmed average	Volume-weighted trimmed average (25%)	Volume-weighted average	Volume-weighted average

2: Classic vs Modern Benchmark Rates

SOFR so far: market data

The market for financial instruments indexed to SOFR is already under development.

1. SOFR Futures (CME)

- 1-month Futures: strip of 7 Futures indexed to arithmetic average of daily SOFR values during the contract delivery month.
- 3-month Futures: strip of 20 Futures indexed to compounded daily SOFR values during the contract reference quarter.

2. SOFR Swaps

Indexed to compounded daily SOFR values during the coupon period (as other OISs), OTC quotes available from brokers

- OTC SOFR OISs: quotes available from brokers (e.g. Tullet, Reuters page USDSROIS=TPSR).
- OTC SOFR vs Fed Fund Basis Swaps: quotes available from brokers (e.g. Tullet, Reuters page USDSRFF=TPSR).
- Only a few transactions observed, liquidity very low at the moment.

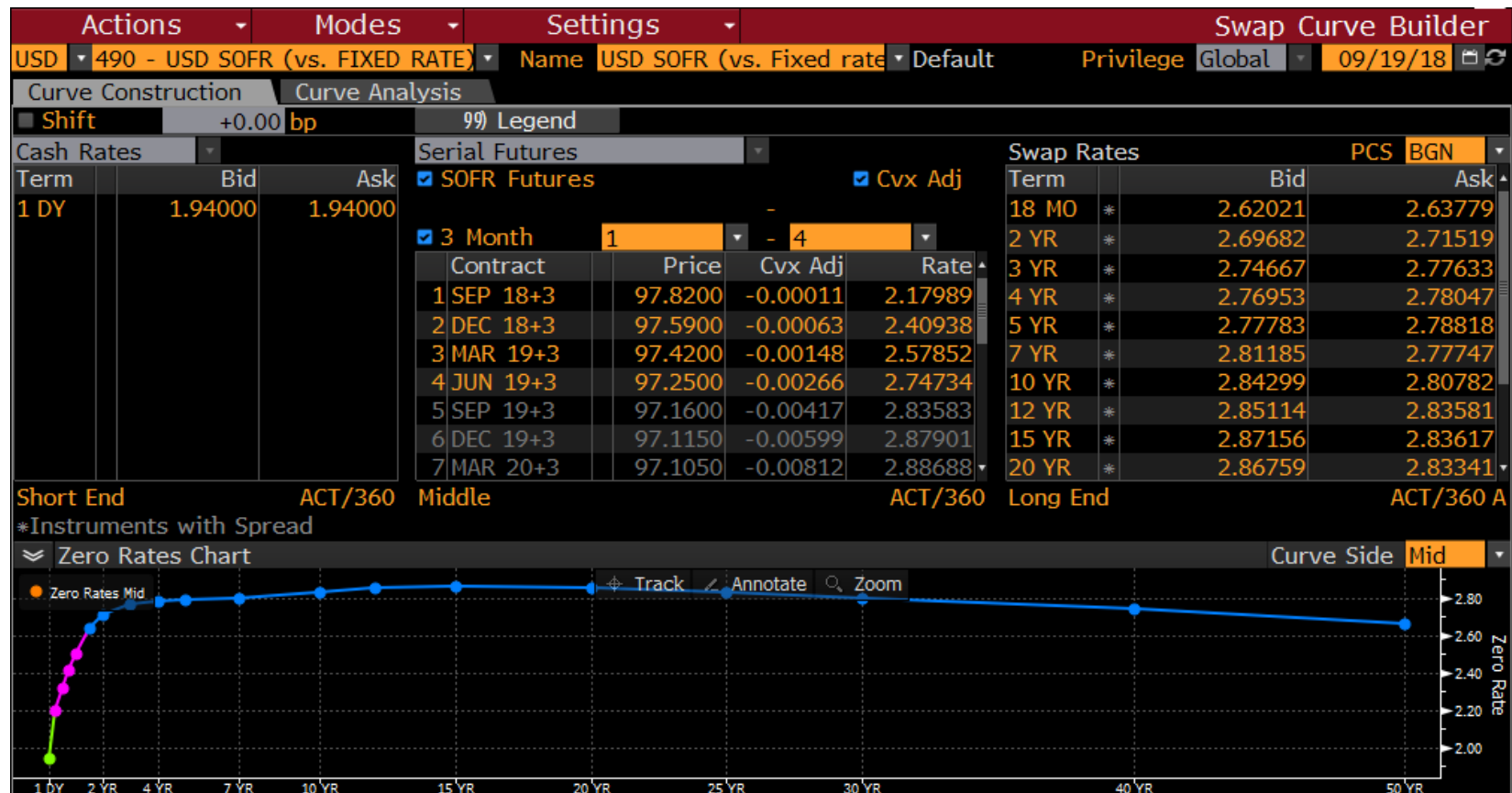
3. SOFR Bonds

Bonds indexed to SOFR have been issued by Banks and Insurances (19 bonds for 19 mld USD as of Oct. 2018).

2: Classic vs Modern Benchmark Rates

SOFR so far: yield curve construction

Bloomberg has developed a USD SOFR yield curve built with 3M SOFR Futures (as SOFR 1M Futures still lack the necessary liquidity) and, for the long end, the existing Fed Funds OIS curve, plus the SOFR vs Fed Fund basis spread curve. Such yield curve can be used to price financial instruments indexed to SOFR.



2: Classic vs Modern Benchmark Rates

SOFR so far: market (il)liquidity

Risk.net

SOFR, so bad: liquidity lags transition ambitions

Thin current trading may lead to poor fallback choices, and dim SOFR's appeal ahead of Libor's death

Robert Mackenzie Smith

04 Feb 2019

NEED TO KNOW

- Trading of SOFR-linked derivatives has been limited since the new rate was launched in April as the US Libor replacement.
- A total of 52 swaps referencing the new rate were traded last year. Open interest in SOFR futures had been growing prior to the end of last year, but stands at less than 1% of the notional for Libor-referencing eurodollar contracts.
- Dealers worry illiquidity could curb the adoption of the new benchmark as the fallback rate in legacy swaps contracts.
- Firms may also be reluctant to transfer Libor positions to SOFR before Libor disappears, which will increase the need for the imperfect solution of fallbacks.
- The present sluggish state of the SOFR market may result in short-sighted responses to Isda's upcoming consultation on the adjustments the new rate will need to act as a fallback.
- Trading in SOFR derivatives should take off once SOFR discounting becomes mandatory; the industry creates a SOFR term rate, kick-starting the cash market; and firms make the required operational changes.

In SOFR interest rate swaps, [first traded](#) in July, the picture is worse. According to [data](#) from swap data repositories, just 52 contracts changed hands in 2018, with a traded notional of \$6.3 billion, while for US dollar Libor-linked interest rate swaps the notional stood at \$111.5 trillion.

Igniting liquidity

So when will SOFR derivatives liquidity improve?

What many participants feel will spark the biggest shift to the new benchmark is the move to SOFR for cashflow discounting and price alignment interest (PAI) – used to calculate interest on posted variation margin – at central counterparties.

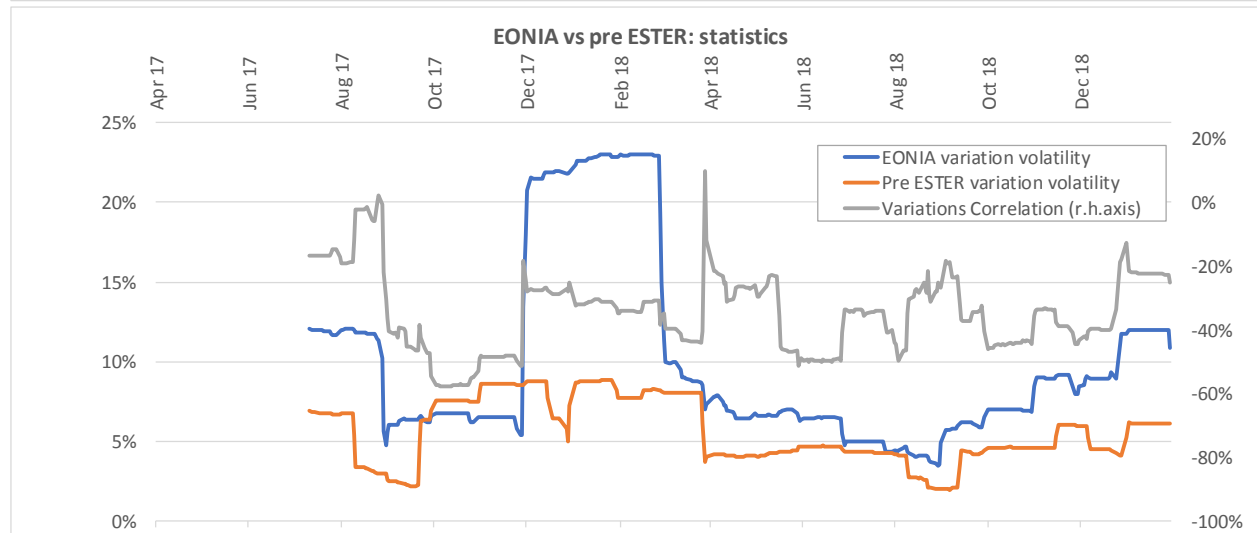
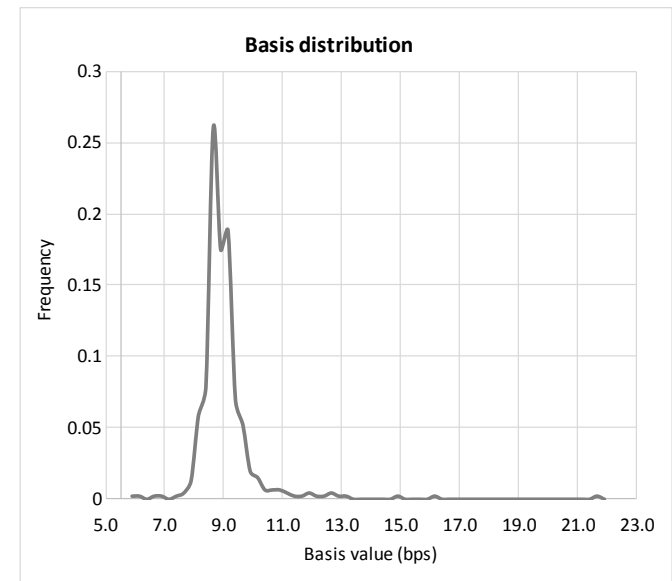
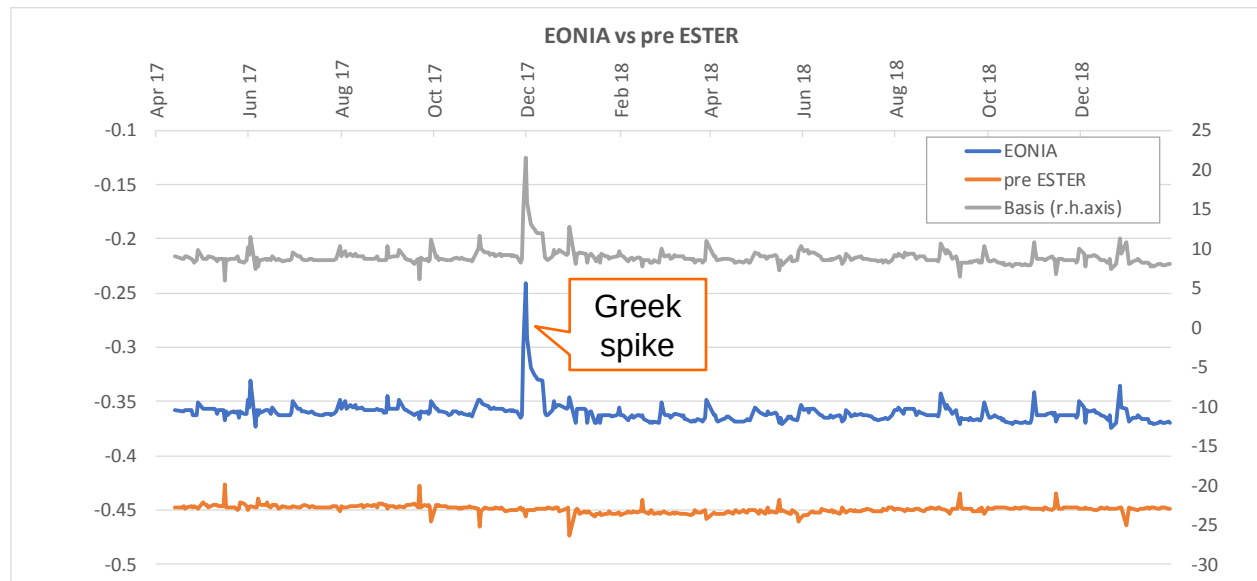
The CME already offers this for SOFR swaps, but SOFR swaps cleared at LCH.Clearnet [still use](#) the effective federal funds rate, as Libor swaps do today.

That should change in the second quarter of 2021 when, according to the ARRC's plan, clearing houses will stop accepting new swap contracts that use the federal funds rate for discounting and PAI, "except for the purpose of closing out or reducing outstanding risk in legacy contracts that use [the federal funds rate]".

Existing contracts using the long-established overnight rate will roll off over time as they mature or are closed out.

2: Classic vs Modern Benchmark Rates

ESTER so far



Basis Distribution Stats		
Min	5.9	bps
Max	21.5	bps
Mean	8.9	bps
Std dev	1.1	bps
Observations	457	

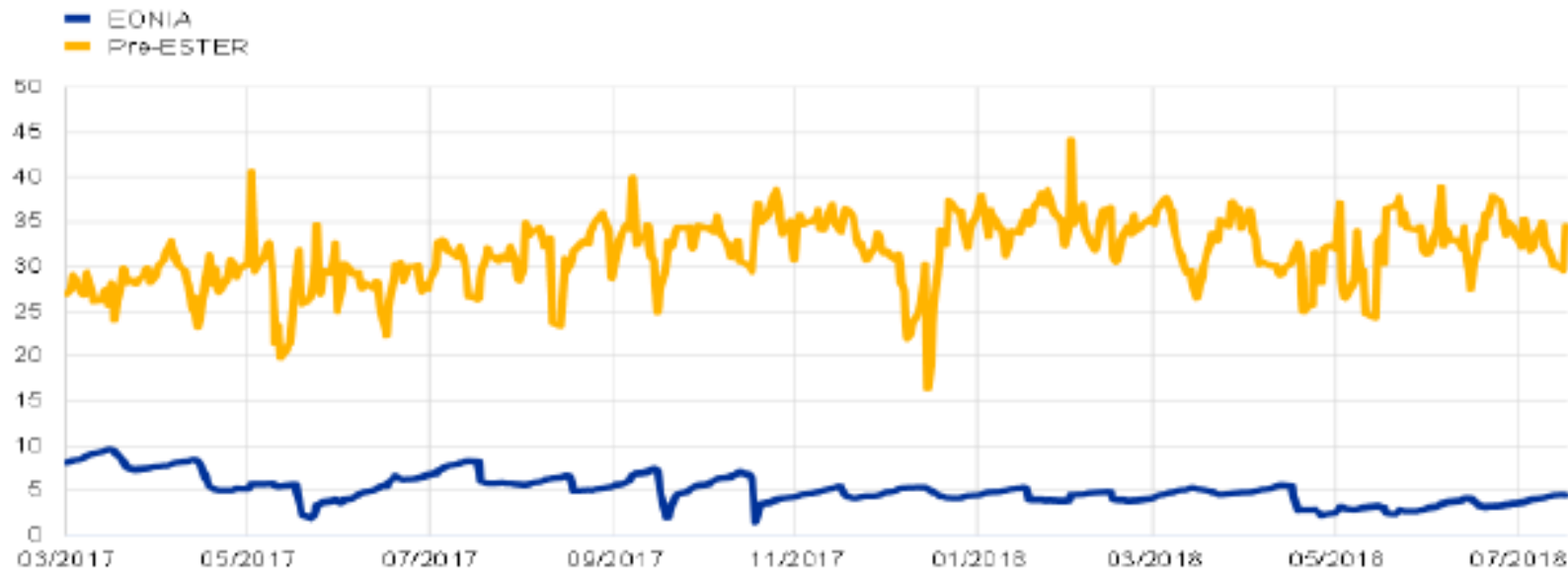
2: Classic vs Modern Benchmark Rates

ESTER so far

Pre-ESTER volumes markedly above those of EONIA

Transaction volumes of EONIA and pre-ESTER

(EUR billion)



Source: Speech by Benoît Coeuré, Member of the Executive Board of the ECB, at the ECB's Money Market Contact Group meeting, Frankfurt am Main, 25 September 2018, Chart 5

The market for financial instruments indexed to ESTER does not exist at the moment.

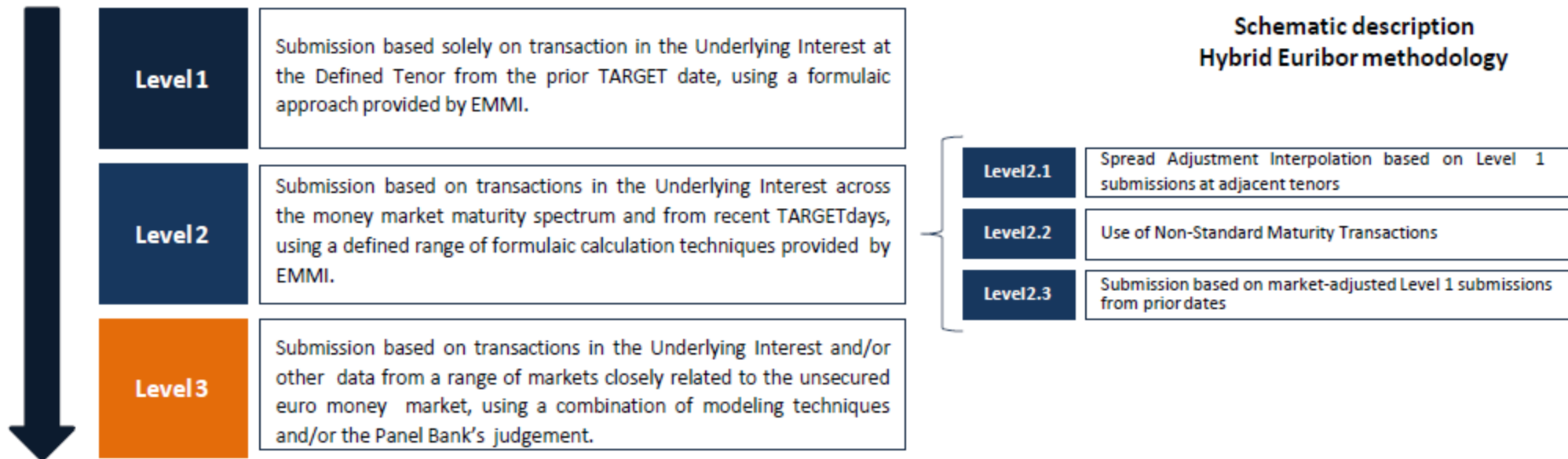


2: Classic vs Modern Benchmark Rates

EURIBOR so far

Hybrid methodology, supported by transactions whenever available, but relies on other techniques or data sources according to input criteria established by EMMI.

Hierarchy: for each day in which the index is calculated, contributing banks will have to base their submissions, for each tenor, on:



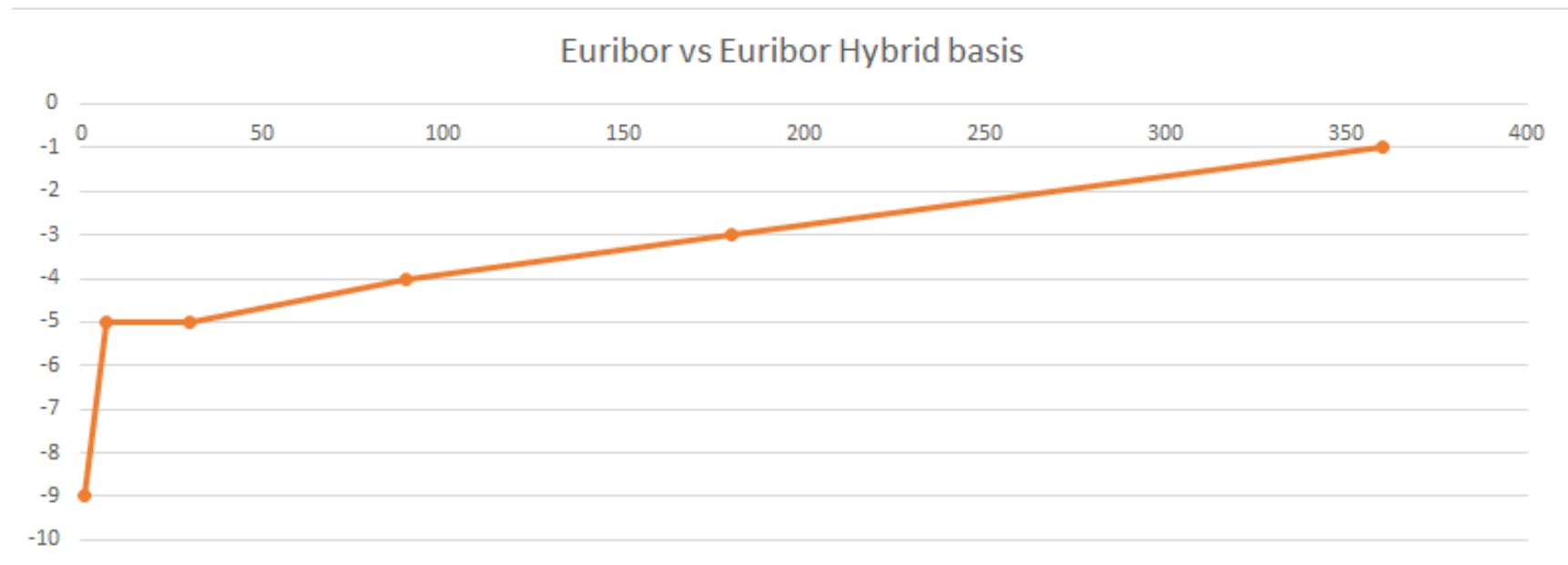
(source: EMMI)

EMMI conducted a test under live conditions from May to July 2018, finding **Hybrid Euribor lower than Euribor by 1-5 bps depending on the tenor 1M, 3M, 6M, 12M.**

2: Classic vs Modern Benchmark Rates

Expected impacts

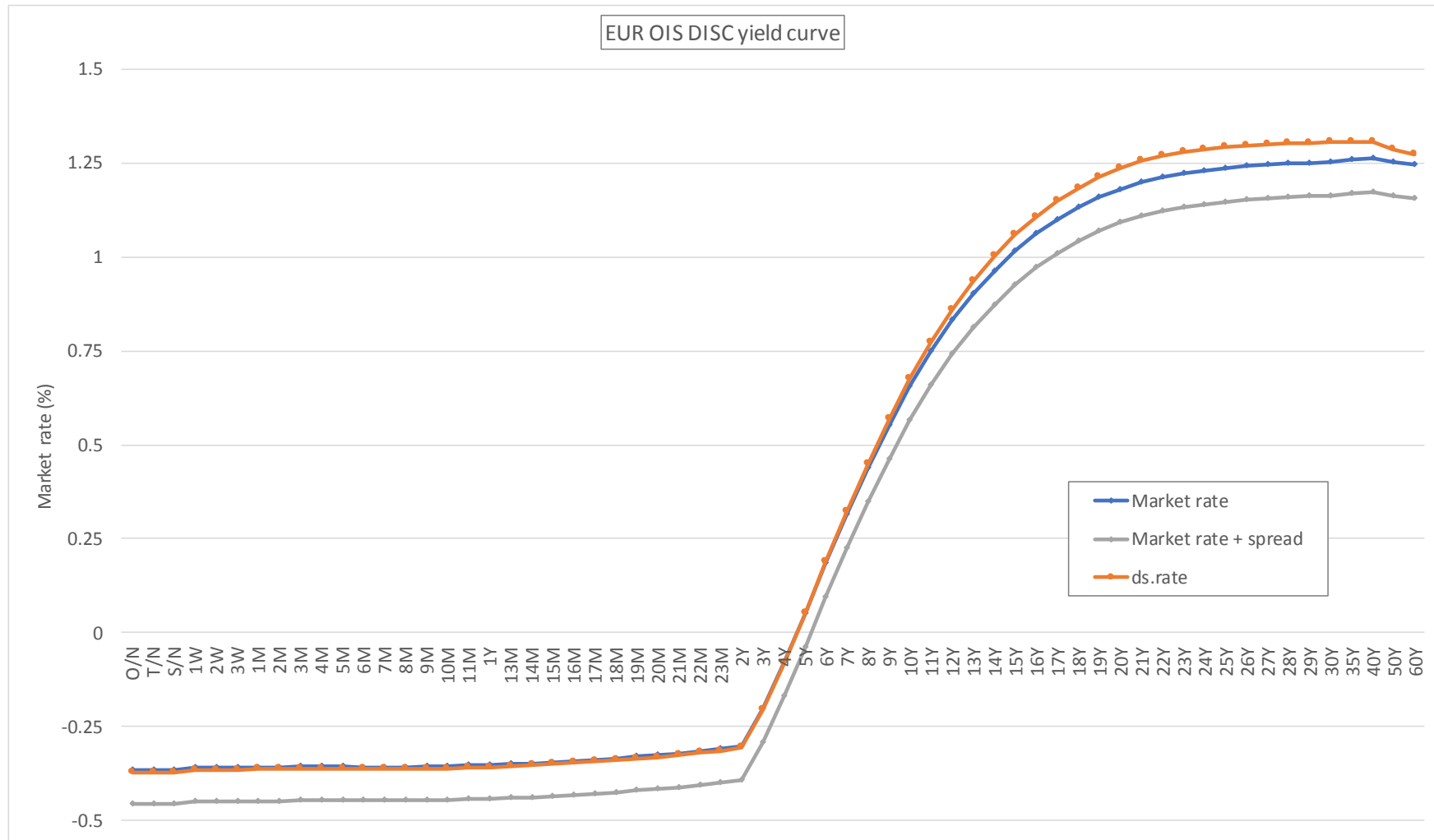
Tenor	Days	Months	Spread (bps)	Spread rounded (bps)	Comment
ON	1	0.033333	-9	-9	Scenario from EONIA - ESTER transition
1W	7	0.233333	-5	-5	Scenario for 1W Euribor tenor (flat 1W-1M)
1M	30	1	-4.739	-5	Scenario for 1M Euribor tenor (flat 1W-1M) and yield curve Euribor 1M
3M	90	3	-4.059	-4	Scenario for 3M Euribor tenor (interpolated) and yield curve Euribor 3M
6M	180	6	-3.040	-3	scenario for 6M Euribor tenor (interpolated) and yield curve Euribor 6M
12M	360	12	-1	-1	scenario for 12M Euribor tenor (interpolated) and yield curve Euribor 12M



Expected impacts of EONIA→ ESTER and EURIBOR→ Hybrid EURBOR transition.

2: Classic vs Modern Benchmark Rates

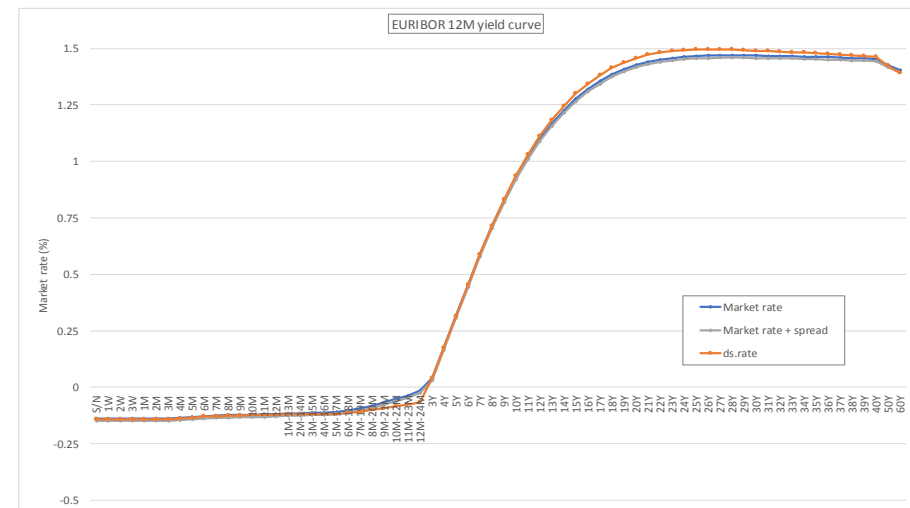
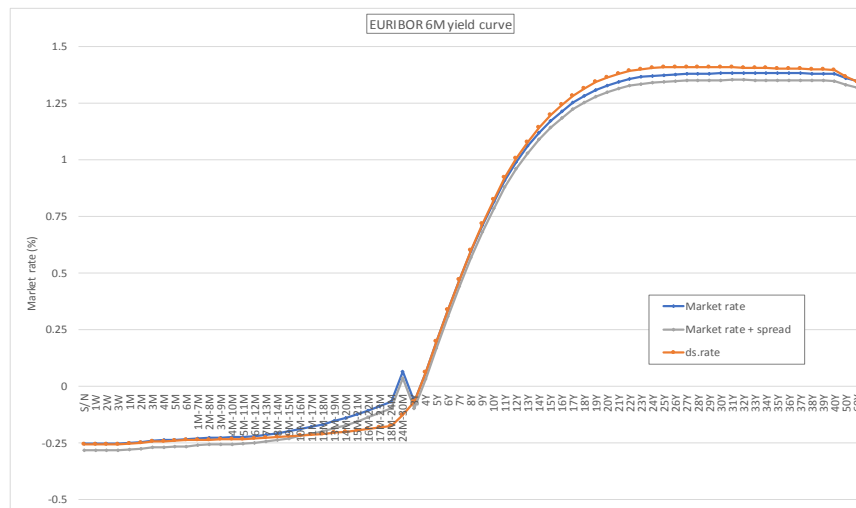
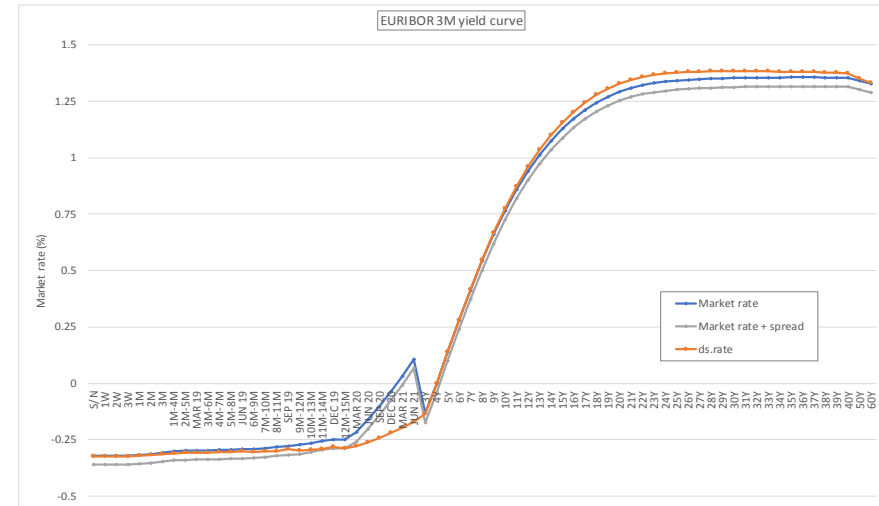
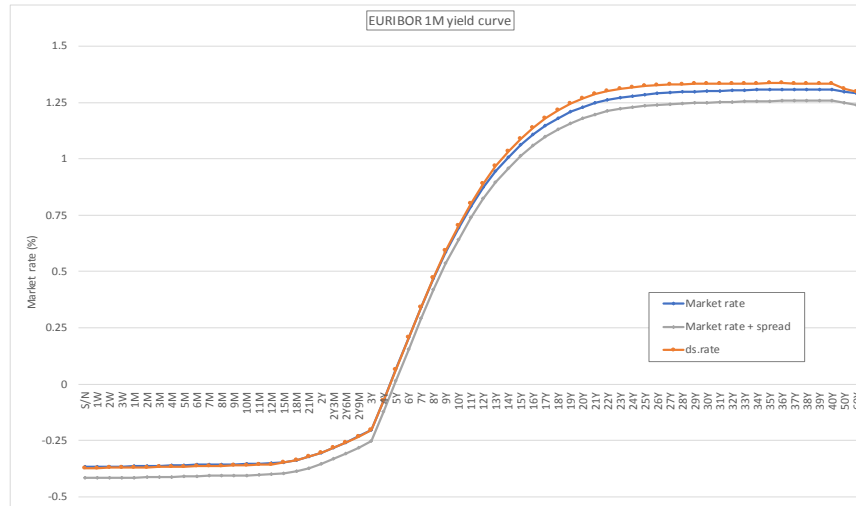
EONIA → ESTER expected impacts



EONIA OIS yield curve as of EoY 2018 (blue), and expected EONIA → ESTER transition impact of -9 bps on market rates (grey) (horizontal scale is non-linear).

2: Classic vs Modern Benchmark Rates

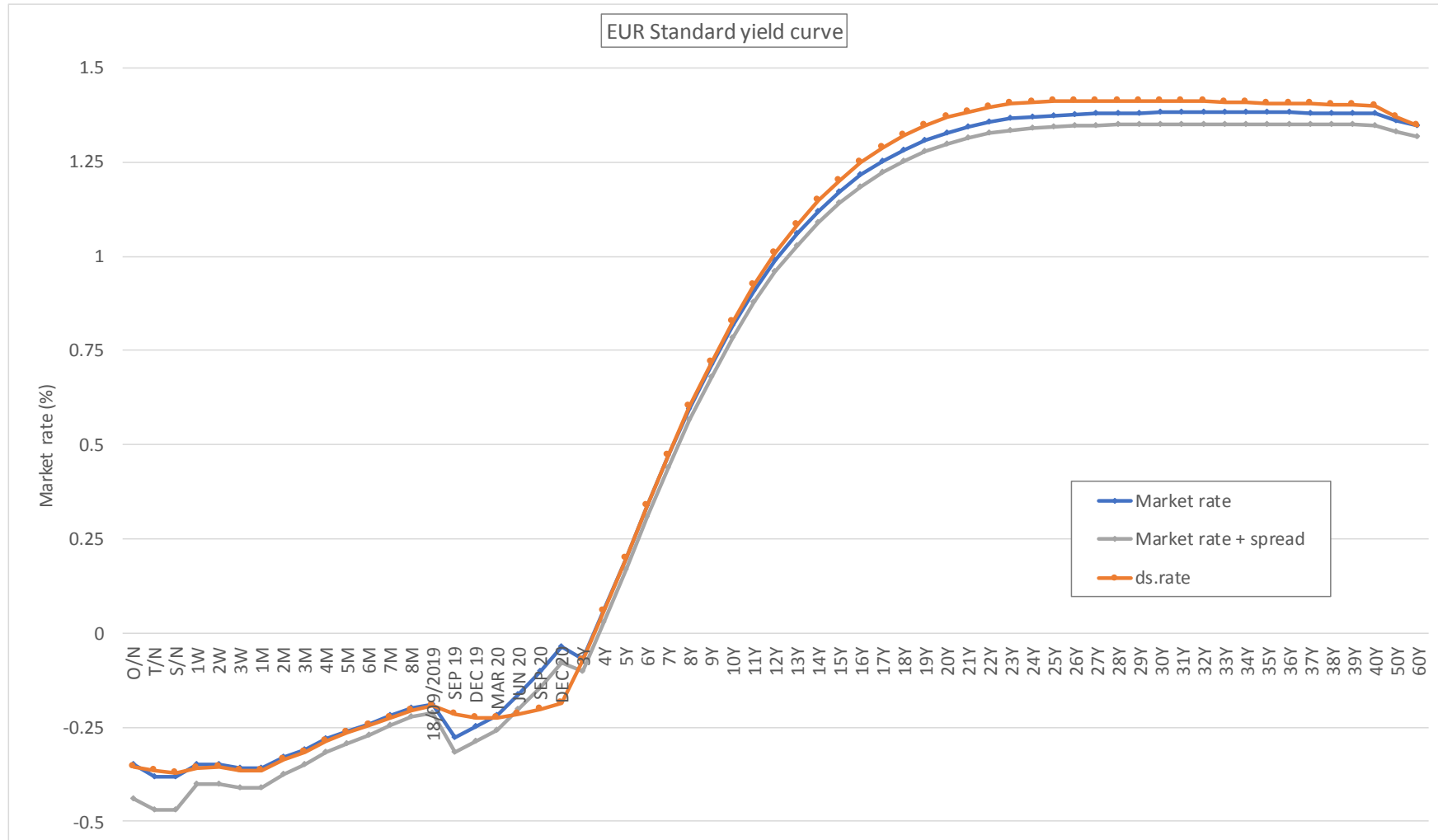
EURIBOR → Hybrid EURIBOR expected impacts



EURIBOR 1M, 3M, 6M, 12M yield curves as of EoY 2018 (blue), and expected EURIBOR → Hybrid EURIBOR transition impact of -5 to -1 bps on market rates (grey) (horizontal scale is non-linear).

2: Classic vs Modern Benchmark Rates

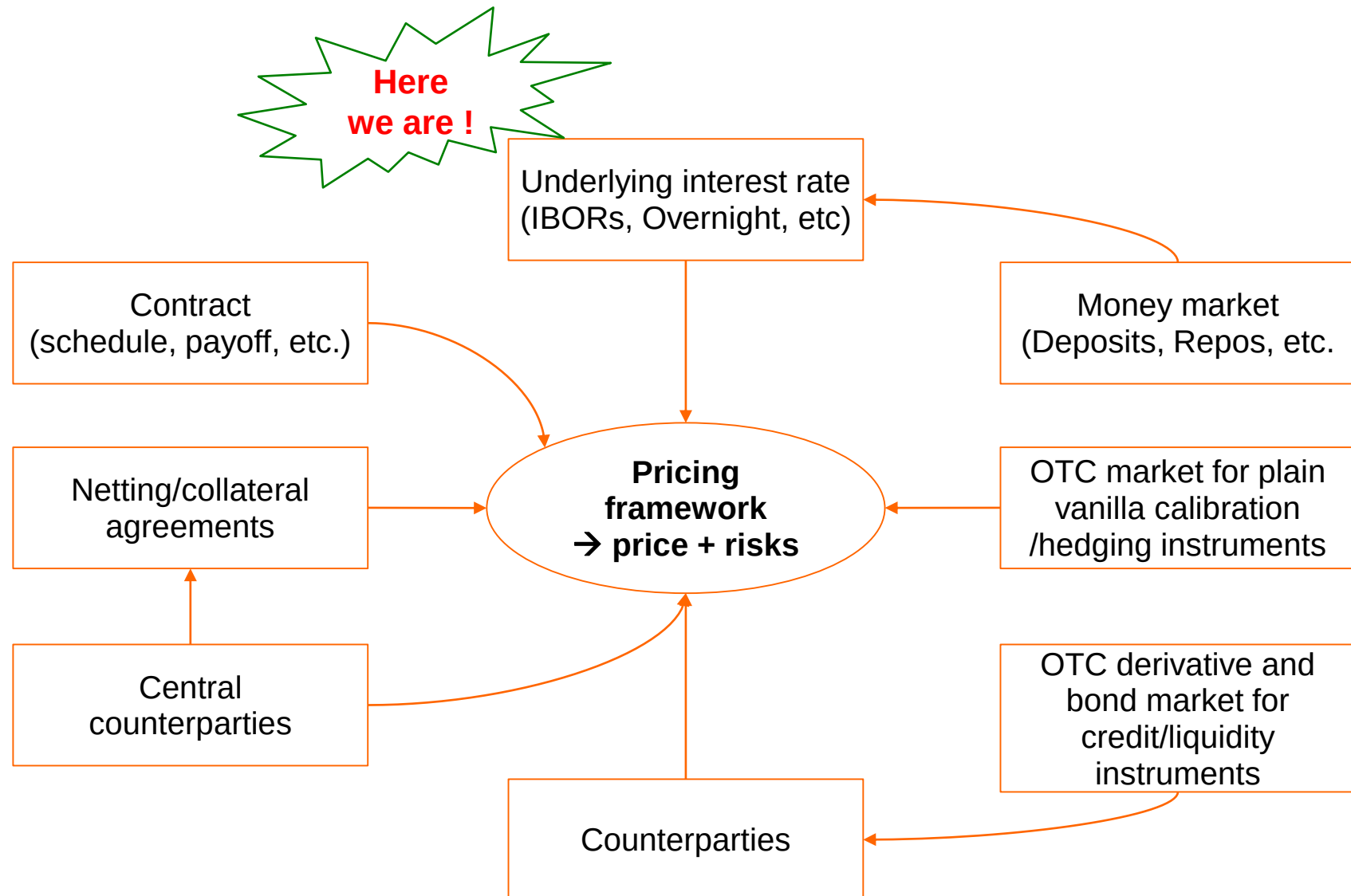
EURIBOR → Hybrid EURIBOR expected impacts



EURIBOR Standard yield curve as of EoY 2018 (blue), and expected EURIBOR → Hybrid EURIBOR transition impact of -5 to -1 bps on market rates (grey) (horizontal scale is non-linear).

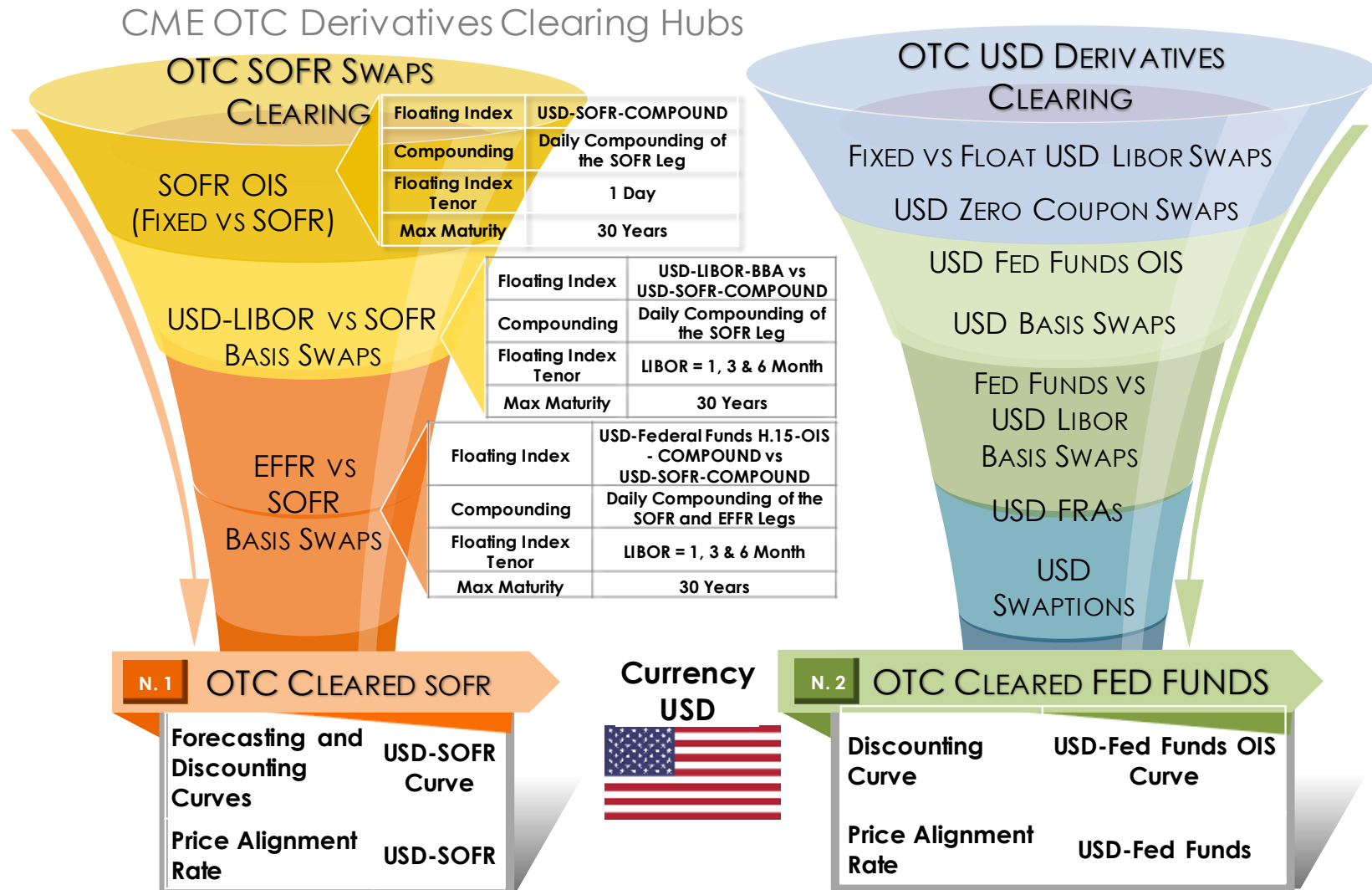
3: Pricing and Risk Management

Pricing financial instruments is a complex task



3: Pricing and Risk Management

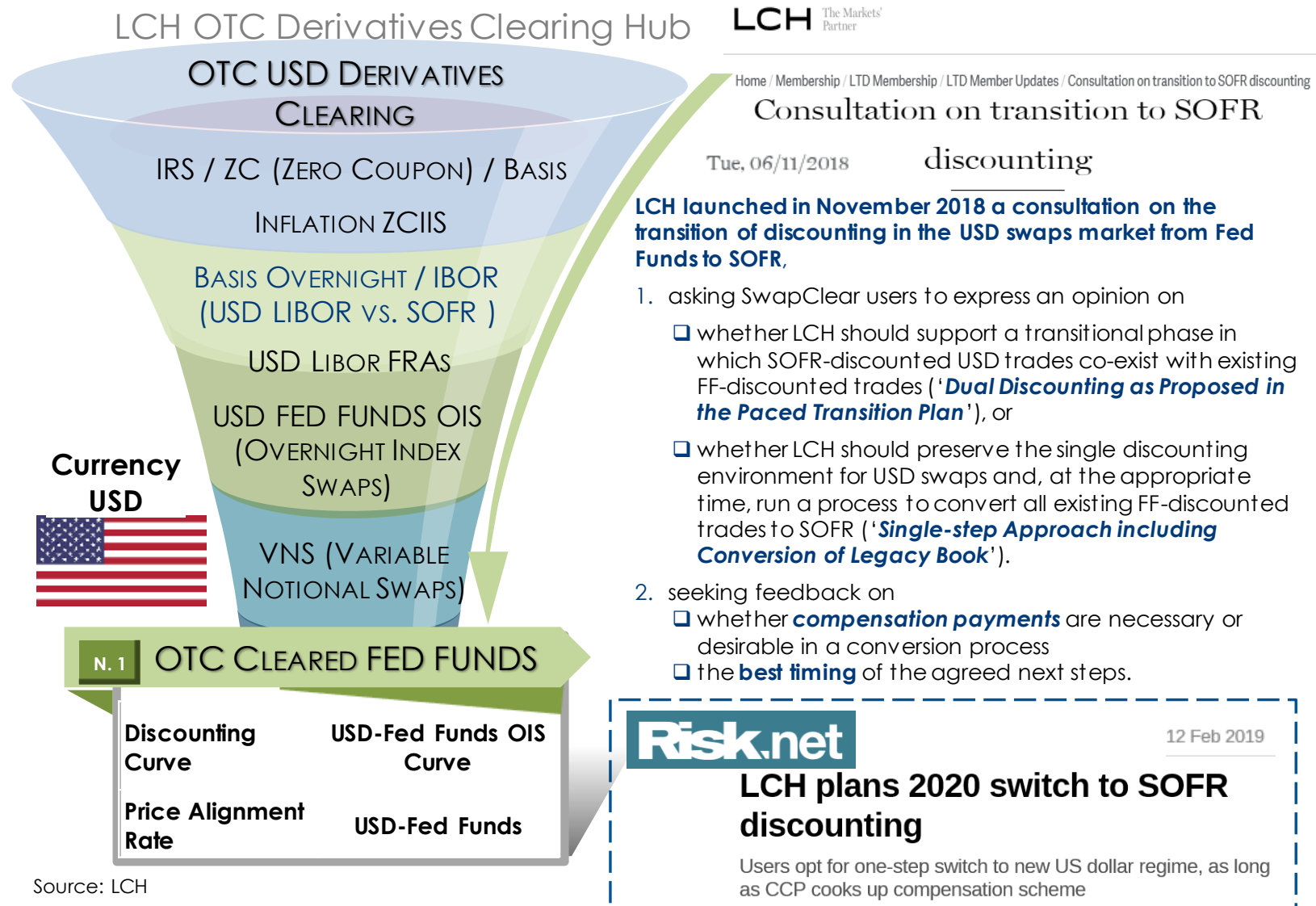
The fundamental role of CCPs: discounting standards



Source: CME Group – OTC SOFR Swaps Clearing 2018

3: Pricing and Risk Management

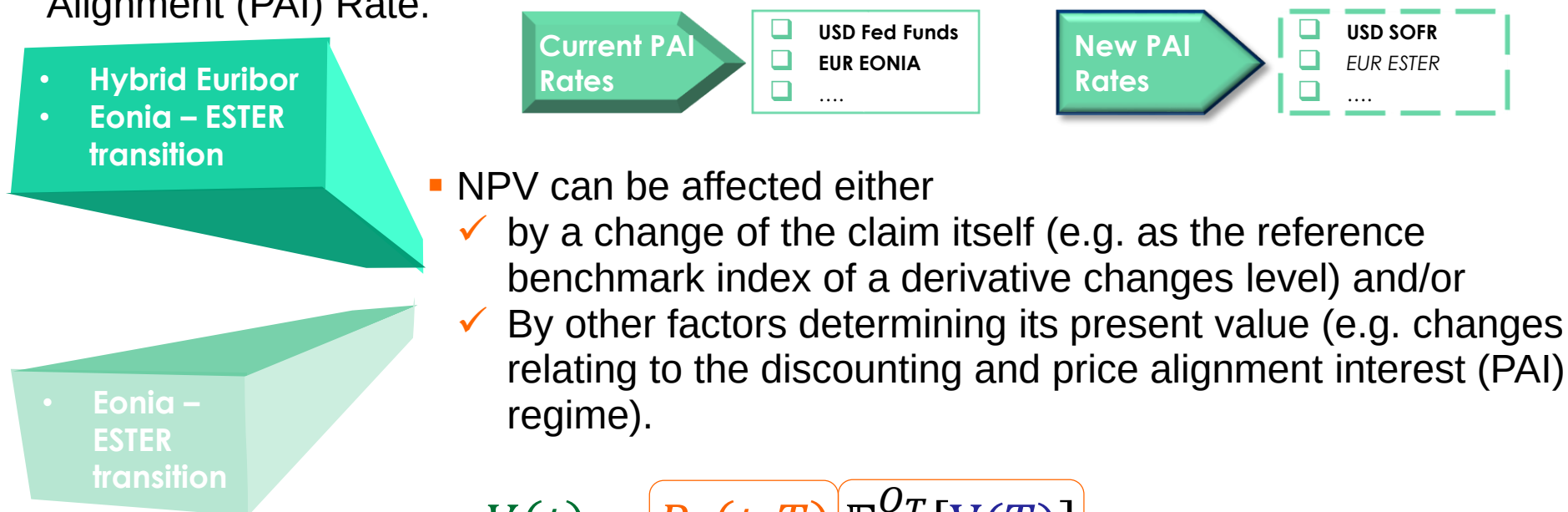
The fundamental role of CCPs: discounting standards



3: Pricing and Risk Management

Pricing impact analysis: overview

- The Risk Free or Base Net Present Value of a derivative as the risk free base component of the derivative Fair Value, not including credit and funding components,
- i.e. as if the derivative were perfectly collateralized by a cash deposit at a given Price Alignment (PAI) Rate.



$$V(t) = P_d(t, T) \mathbb{E}_t^{Q^T} [V(T)]$$

Source of impacts in the risk neutral valuation of a general instruments under IR benchmark transition

EONIA → ESTER
discounting

- EURIBOR → Hybrid EURIBOR forwarding
- EONIA → ESTER forwarding
- Volatilities
- ...

3: Pricing and Risk Management

Pricing impact analysis: transition plans

- Transition Plans (TPs) providing a (non-zero) spread between EONIA and ESTER aim to keep projected cash flow curve levels largely unchanged or at least to minimise the impact.
- Discounting and PAI regime are closely related and should be treated consistently.
 - TPs which involve a bifurcation into benchmarks instruments under multiple, parallel discounting and PAI regimes introduce the risk of pricing in an incomplete market, especially in cases where EONIA and ESTER were independent benchmarks and where the transition were under significant time pressure due to regulatory constraints like the EU Benchmarks Regulation (BMR).
 - If EONIA and ESTER were dependent benchmarks but operated in parallel discounting regimes, the proxy risk would be less prominent, if not completely avoided.
 - In all other cases, the liquidity will always be focused on a single discounting regime and a set of benchmark instruments resulting in a low perceived impact.

3: Pricing and Risk Management

Pricing impact analysis: interest rate [1/3]

The interest rate curves are obtained as follows.

- **Single currency**: yield curve construction is based on bootstrapping algorithm which obtains the curve pillars recursively from quoted financial instruments, using the pricing formulas of collateralized instruments (OIS discounting).
- **Cross currency**: cross currency basis yield curves construction is based on the same methodology above, using quoted Cross Currency Swaps and their corresponding pricing formulas in the base collateral currency.
- **CCP Basis** (e.g. EUREX-LCH): basis yield curve construction is based on the same methodology above, using the quoted Swaps cleared to each CCPs.
- **Funding curves**: funding curves construction is based on a fitting algorithm of quoted benchmark or comparable Bonds prices using e.g. Nelson-Siegel or Svensson model.

Because of the consistent usage of OIS discounting, **the bootstrap algorithm is intrinsically multi-curve**. As a consequence a change of EONIA discount rates also affects EURIBOR forward rates.

BTW, the impact of EONIA → ESTER transition on forward rates **is negligible**.

3: Pricing and Risk Management

Pricing impact analysis: interest rate [2/3]

Accordingly, we may expect the following impacts on yield curves.

Expected impacts on yield curves		As is	EONIA → ESTER	EURIBOR → Hybrid EURIBOR
Discount curve	Forwards	EONIA OIS	ESTER OIS	EONIA OIS
	Discounts	EONIA OIS	ESTER OIS	EONIA OIS
Forward curves	Forwards	EURIBOR IRS	EURIBOR IRS	Hybrid EURIBOR IRS
	Discounts	EONIA OIS	ESTER OIS	ESTER OIS
EUR CCS basis curves (e.g. EURUSD)	Forwards	EURIBOR IRS USDLIBOR IRS	EURIBOR IRS USDLIBOR IRS CCS EURUSD basis	Hybrid EURIBOR IRS USDLIBOR IRS CCS EURUSD basis
	Discounts	EONIA OIS EONIA CCS basis	ESTER OIS ESTER CCS Basis	ESTER OIS ESTER CCS Basis
Funding curves	Discounts	Benchmark bond prices	Benchmark bond prices	Benchmark bond prices

Red = impacted quantities.

3: Pricing and Risk Management

Pricing impact analysis: interest rate [3/3]

Accordingly, we may expect the following impacts on IR financial instruments

Expected impacts on linear IR financial instruments	EONIA → ESTER	EURIBOR → Hybrid EURIBOR
IR Derivatives (unfunded)	▪ Forward Rate Agreements (FRA) ▪ Overnight Indexed Swaps (OIS) ▪ Swaps (IRS) ▪ Basis Swaps (BIRS) ▪ Cross Currency Swaps (CCS)	▪ Forward Rate Agreements (FRA) ▪ Overnight Indexed Swaps (OIS) ▪ Swaps (IRS) ▪ Basis Swaps (BIRS) ▪ Cross Currency Swaps (CCS)
IR Funded instruments	▪ Deposits ▪ Repos ▪ Loans, mortgages ▪ Bonds	▪ Deposits ▪ Repos ▪ Loans, mortgages ▪ Bonds (floating rate) ▪ Bonds (fixed rate)

Red = impacted quantities.

Bonds and other IR funded instruments suffer no impacts under the EONIA → ESTER transition according to the assumption that the funding rate, defined as the sum of a reference rate (e.g. EURibor3M) and a credit spread, will remain unchanged. The decrease of EURIBOR is compensated by a corresponding increase of the spread, keeping the funding spread constant.

3: Pricing and Risk Management

Pricing impact analysis: interest rate Cap/Floor volatility [1/3]

The Cap/Floor volatility surfaces (expiry x strike) are obtained from market quotes as follows:

- **EURIBOR tenors 3M, 6M**: are obtained from market quotes (either in terms of term volatilities or premia) and mapped in terms of **implied forward shifted-lognormal (or normal) volatility surfaces** (i.e. Caplet/Floorlet volatilities) using OIS discounting;
- **EURIBOR tenors 1M, 12M**: are obtained from the surfaces above via Kienitz model using OIS discounting.
- In both cases a **SABR model** is calibrated and used to obtain non-quoted expiries/strikes.

3: Pricing and Risk Management

Pricing impact analysis: interest rate Cap/Floor volatility [2/3]

Accordingly, there are two possible alternative scenarios for the EONIA/ESTER transition:

- **Scenario 1 (Constant Term Volatility)**: keep constant the market term volatility surface → use ESTER OIS discounting to obtain premia and forward volatilities;
- **Scenario 2 (Constant Premium)**: keep constant the market term volatility surface and the premia (obtained using EONIA OIS discounting) → use ESTER OIS discounting to obtain forward volatilities.

Expected impacts of EONIA → ESTER	As is	Scenario 1: Constant Term Volatility	Scenario 2: Constant Premium
Term volatility (normal)	Market quotes	Market quotes	Market quotes
Cap/Floor premium	Black formula (EONIA OIS discounting)	Black formula (ESTER OIS discounting)	Black formula (EONIA OIS discounting)
Forward volatility	Inverted Black formula (EONIA OIS discounting)	Inverted Black formula (ESTER OIS discounting)	Inverted Black formula (ESTER OIS discounting)

3: Pricing and Risk Management

Pricing impact analysis: interest rate Cap/Floor volatility [3/3]

Accordingly, we may expect the following impacts on market data and financial instruments.

Expected impacts of EONIA → ESTER	Scenario 1: Constant Term Volatility	Scenario 2: Constant Premium
Impacted market data	▪ Term volatilities ▪ Cap/Floor premia ▪ Forward volatilities	▪ Term volatilities ▪ Cap/Floor premia ▪ Forward volatilities
Impacted financial instruments	▪ Caps/Floors plain vanilla ▪ Caps/Floors digital ▪ Any structured product using Cap/Floor volatilities, e.g. Ratchet/Sticky Caps/Floors ▪ Any product involving an EURIBOR convexity adjustment (e.g. in arrear IRSs).	

What about EURIBOR → Hybrid EURIBOR transition ?

- EURIBOR forward changes
- EURIBOR volatility remains essentially constant → Constant Term Volatility
- Caps/Floor prices change.

3: Pricing and Risk Management

Pricing impact analysis: interest rate Swaption volatility

The swaption volatility cube (option expiry x swap tenor x option strike) are obtained from market quotes as follows:

- **Swaption volatility cube for EURIBOR tenor 6M**: is obtained from market **forward premia** (physical and cash delivery) for ATM swaption, from shifted-lognormal implied volatility spread for OTM swaptions, and mapped in terms of **implied forward shifted-lognormal volatility cubes**.
- **Swaption volatility cube for other EURIBOR tenors** (i.e. 3M): is obtained from the cube above using par basis point volatility (i.e. equating forward swap rate variances) and ESTER OIS discounting.
- In both cases a **SABR model** is calibrated and used to obtain non-quoted expiries/tenors/strikes.

Since the impacts of EONIA → ESTER transition on EURIBOR forward rates are negligible, we may conclude that:

- the impact on par swap rates are negligible,
- the impact on Swaption implied forward shifted-lognormal volatility cubes are negligible,
- the impact on prices of financial instruments using such volatilities (CMS, CMS Spread Options, Bermudan Swaptions, some exotic) will amount to a pure discounting effect → **there is no need to build swaption volatility scenarios**.

3: Pricing and Risk Management

Pricing impact analysis: other financial instruments

The

- Interest rates derivatives:
 - ✓ IR exotics: priced using IR models (i.e. Short Rate Models, Libor Market model)
- Other derivatives (equity/FX/credit/commodity):
 - ✓ Impacts due to discounting effects (both in case of OIS and Standard discounting)
 - ✓ Impacts due to implied volatilities, similarly to IR Caps/Floors
- Certificates:
 - ✓ Expected impacts on Certificates pricing are negligible compared to the market bid ask spreads, and due to compensations between the input credit curve (CERTUNICA) and the discounting curve.
- XVAs: see next slides

3: Pricing and Risk Management

Risk Management Issues

Banking Book Interest Rate Risk

- Analysis of the expected changes in the shift sensitivities – ΔEVE .
- Analysis of expected changes in the Net Interest Income.
- Impact Analysis on hedges in the IAS/IFRS framework (Hedge Accounting), both for assets (Credits and Bond Portfolios) and liabilities (Issued Bonds) especially with respect to the Effectiveness Test.

Liquidity Risk

- Analysis of the expected changes in the shift sensitivities – ΔEVE .
- Analysis of expected changes in the Net Interest Income.
- Impact Analysis on hedges in the IAS/IFRS framework (Hedge Accounting), both for assets (Credits and Bond Portfolios) and liabilities (Issued Bonds) especially with respect to the Effectiveness Test.



Operational Risk

- Impact analysis on the revision of internal processes and procedures
- Early involvement of technical and control functions via a project organization may provide an oversight on operative decisions and address potential issues since their discovery, in order to minimize ICT, model and legal risks.

Market Risk

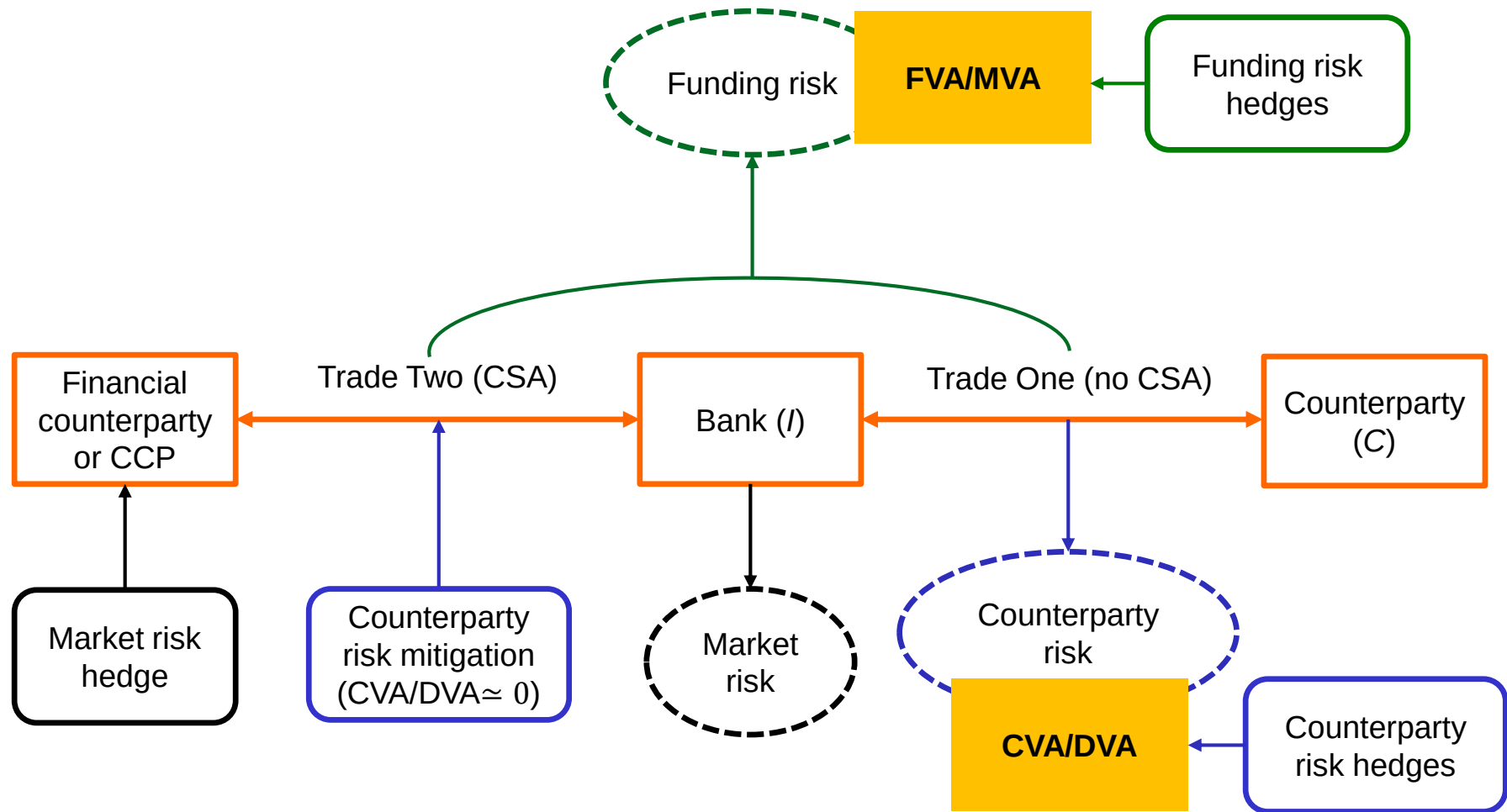
- Analysis of the changes in sensitivities and VaR limits.
- Impact analysis with respect to Internal Model Market Risk Capital Requirements.
- Analysis of new risk factors, construction of new time series.
- Analysis of a potential Risk Not In Model capital requirement under the current framework, and a Non-Modellable Risk Factor capital add-on under the FRTB.

Counterparty Credit Risk

- Impact analysis with respect to Internal Model Counterparty Risk Capital Requirements.
- Analysis of time simulation of new risk factors and their impact on counterparty future exposures.
- Impact assessment on credit line absorption and internal limits on Credit Portfolio and CVA VaR

4: Focus on XVAs

Overview



4: Focus on XVAs

The goal

Proposition: **under the EONIA-ESTER transition, the mark to market change is compensated by the FVA.**

We will proof the proposition above in a simplified situation, as follows.

- We consider a single uncollateralized trade with a single future cashflow $\mathcal{C} > 0$ received by the investor I from the counterparty C .
- → The Debt Valuation Adjustment (DVA) for I is zero.
- We suppose that the counterparty is default-free → also the Credit Valuation Adjustment (CVA) for I is zero.
- We define the base value $V^0(t)$ of the trade as the value under perfect collateral, without any counterparty and funding risk,

$$V^0(t) = P_d(t, T) \mathcal{C}$$

- The fair value of the trade, with the usual OIS discounting and including XVAs, is given by

$$V(t) = V^0(t) + \text{FVA}(t) = P_d(t, T) \mathcal{C} + \text{FVA}(t)$$

4: Focus on XVAs

Defaultable Zero Coupon Bond [1]

- Let's consider the probabilistic framework $(\Omega, \mathcal{F}, \mathcal{F}_t, \mathbb{Q})$ with the bank account numeraire B_t which evolves at the risk free rate $r(t)$, with $\mathcal{F}_t$ market filtration.
- We define the discount factor and the zero coupon bond

$$D(t, T) = \frac{B(t)}{B(T)} = e^{-\int_t^T r(s)ds}$$

$$P_d(t, T) = \mathbb{E}_t^{\mathbb{Q}}[D(t, T)1] = \mathbb{E}_t^{\mathbb{Q}}[e^{-\int_t^T r(s)ds}].$$

- We define the risky zero coupon bond with no recovery rate the bond which guarantees a unitary payment at maturity if the issuer I is not defaulted before,

$$P_I(t, T) = \mathbb{E}_t^{\mathbb{Q}}[D(t, T)\mathbb{1}_{\{\tau_I > T\}}]$$

- Supposing that interest rates and default time are independent, we obtain

$$P_I(t, T) = \mathbb{E}_t^{\mathbb{Q}}[e^{-\int_t^T r(s) + s_I(s)ds}]$$

4: Focus on XVAs

Defaultable Zero Coupon Bond [2]

- Let's now consider the more realistic case where recovery rate is not zero. We identify it with the process $\{\mathcal{R}(t)\}_{t \geq 0}$. It can be shown that the expression of the price of a risky zero coupon bond becomes (for simplicity $\tau_I > t$)

$$\begin{aligned} P_I(t, T) &= \mathbb{E}_t^{\mathbb{Q}} \left[e^{-\int_t^{\tau_I} r(s) ds} \mathcal{R}(\tau_I) \mathbb{1}_{\{\tau_I \leq T\}} + e^{-\int_t^T r(s) ds} \mathbb{1}_{\{\tau_I > T\}} \right] \\ &= \mathbb{E}_t^{\mathbb{Q}} \left[\int_t^T \mathcal{R}(u) \gamma(u) e^{\int_t^u (r(s) + \gamma(s)) ds} du \right] + \mathbb{E}_t^{\mathbb{Q}} \left[e^{-\int_t^T (r(s) + \gamma(s)) ds} \right] \end{aligned}$$

- For the recovery rate we adopt the «recovery of treasury» model, such that

$$\mathcal{R}(t) = R P_d(t, T) \quad 0 \leq R \leq 1$$

- Under such model the price of the risky zero coupon bond simplifies as

$$P_I(t, T) = P_d(t, T) (1 - (1 - R)(1 - S_I(t, T)))$$

where $S_I(t, T) := \mathbb{E}_t^{\mathbb{Q}}[\mathbb{1}_{\{\tau_I > T\}}] = \mathbb{Q}(\tau_I > T)$ is the survival probability of I until time T , evaluated in t .

4: Focus on XVAs

XVA impact [1]

- We define the Funding Valuation Adjustment (FVA) intended as a Funding Cost

$$\text{FVA}(t) = -\mathbb{E}_t^{\mathbb{Q}} \left\{ \int_t^T D(t, u) \left[\sum_{c=1}^N H_c(u) \mathbb{1}_{\{\tau_C > u\}} \right]^+ s_I(u) \mathbb{1}_{\{\tau_I > u\}} du \right\}$$

where $H_c(u)$ is the total exposure relative to the counterparty c in the funding set $\{1, \dots, N\}$ and $s_I(t)$ is the instantaneous funding spread for I .

- For a single positive cashflow and no counterparty risk, the FVA reduces to

$$\begin{aligned} \text{FVA}(t) &= -\mathbb{E}_t^{\mathbb{Q}} \left\{ \int_t^T D(t, u) [\mathbb{E}_u^{\mathbb{Q}} [D(u, T)C]]^+ s_I(u) \mathbb{1}_{\{\tau_I > u\}} du \right\} \\ &= -C \mathbb{E}_t^{\mathbb{Q}} \left\{ \int_t^T D(t, u) [P_d(u, T)]^+ s_I(u) \mathbb{1}_{\{\tau_I > u\}} du \right\} \\ &= -C \int_t^T \mathbb{E}_t^{\mathbb{Q}} \{ D(t, u) P_d(u, T) s_I(u) \mathbb{1}_{\{\tau_I > u\}} \} du \\ &= -C \int_t^T s_I(u) P_d(u, T) \mathbb{E}_t^{\mathbb{Q}} \{ D(t, u) \} \mathbb{E}_t^{\mathbb{Q}} \{ \mathbb{1}_{\{\tau_I > u\}} \} du \\ &= -C \int_t^T s_I(u) P_d(u, T) P_d(t, u) S_I(t, u) du \end{aligned}$$

4: Focus on XVAs

XVA impact [2]

- If we observe that $P_d(t, u)P_d(u, T) = P_d(t, T) \quad t \leq u \leq T$

we get
$$\text{FVA}(t) = -\mathcal{C}P_d(t, T) \int_t^T s_I(u)S_I(t, u)du$$

- Using the following relationship (see appendix)

$$s_I(u) = \gamma_I(t)(1 - R) = -\frac{\partial_u S_I(t, u)}{S_I(t, u)}(1 - R)$$

where $\gamma_I(t)$ is the hazard rate, we obtain

$$\begin{aligned}\text{FVA}(t) &= -(1 - R)\mathcal{C}P_d(t, T) \int_t^T \left[-\frac{\partial_u S_I(t, u)}{S_I(t, u)} \right] S_I(t, u)du \\ &= (1 - R)\mathcal{C}P_d(t, T) \int_t^T \partial_u S_I(t, u)du \\ &= (1 - R)\mathcal{C}P_d(t, T)[S_I(t, T) - 1] \\ &= \mathcal{C}P_d(t, T)[R + S_I(t, T) - 1 - RS_I(t, T)] \\ &= \mathcal{C}P_d(t, T)[R + S_I(t, T)(1 - R) - 1] \\ &= \mathcal{C}[R + P_d(t, T)S_I(t, T)(1 - R) - P_d(t, T)] \\ &= -\mathcal{C}[P_d(t, T) - P_I(t, T)]\end{aligned}$$

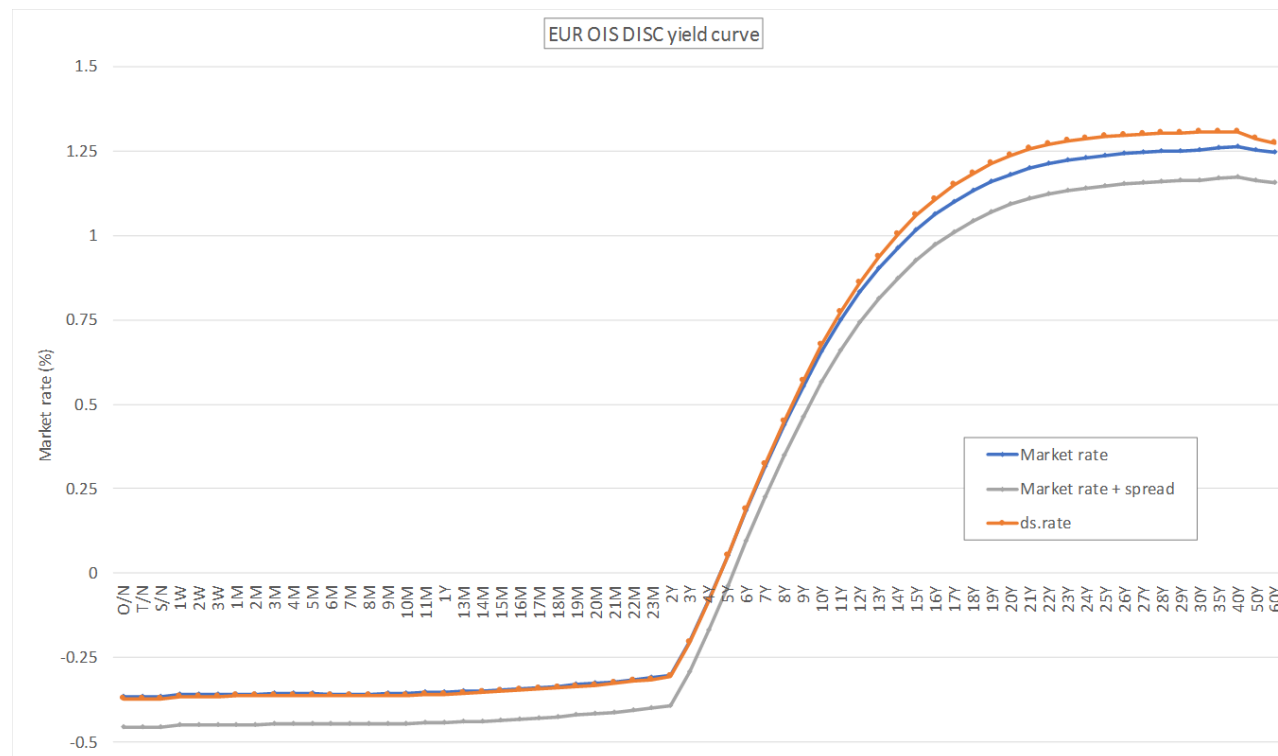
4: Focus on XVAs

XVA impact [3]

- We now assume that the ESTER OIS curve is given by the EONIA OIS curve minus a constant spread (-9 bps at the moment),

$$R_d^{EST}(t, T) = R_d^{EON}(t, T) - \Delta \quad \forall T > t \quad \Delta > 0$$

$$R_x(t, T) := -\frac{1}{\tau(t, T)} \log P_x(t, T)$$



4: Focus on XVAs

XVA impact [4]

- Under the EONIA → ESTER transition the following quantities are impacted (the new value is denoted by «'»)

- ✓ Discount factors

$$P_d(t, T) \rightarrow P'_d(t, T)$$

- ✓ Funding (Zero) Rate of the risky ZCB

$$R_I(t, T) := R_d^{EON}(t, T) + \mathcal{S}_I(t, T) \rightarrow R_d^{EST}(t, T) + \mathcal{S}'_I(t, T) := R'_I(t, T)$$

$$\mathcal{S}_I(t, T) = \frac{1}{\tau(T - t)} \log \frac{P_d(t, T)}{P_I(t, T)},$$

- Assuming that the transition does not affect the overall ZCB value, then

$$\begin{aligned} P'_I(t, T) &= P_I(t, T) \\ \Rightarrow R'_I(t, T) &= R_I(t, T) \\ \Rightarrow \mathcal{S}'_I(t, T) &= \mathcal{S}_I(t, T) + \Delta \end{aligned}$$

4: Focus on XVAs

XVA impact [5]

- Putting all the pieces into the equation of fair value of the trade, we obtain finally

$$V(t) = V^0(t) + \text{FVA}(t) = P_d(t, T) \mathcal{C} + \text{FVA}(t)$$

$$\begin{aligned} V'(t) &= \mathcal{C} P'_d(t, T) - \mathcal{C} [P'_d(t, T) - P'_I(t, T)] \\ &= \mathcal{C} P'_I(t, T) \\ &= \mathcal{C} P_I(t, T) \\ &= V(t). \end{aligned}$$

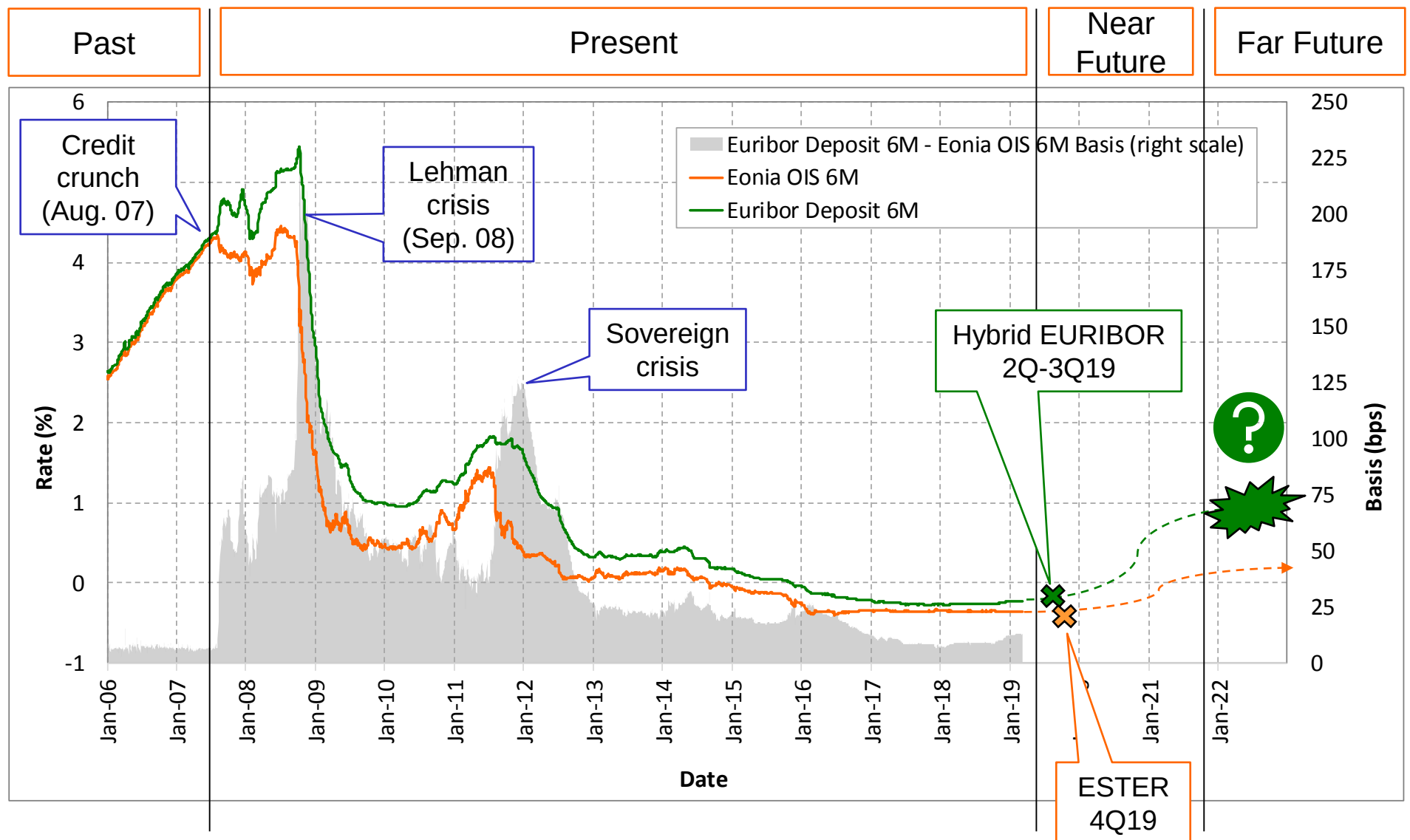
c.v.d.

We have proved that, under appropriate hypotheses, the impact of EONIA → ESTER transition on a trade is neutralised by the corresponding variation of the funding spread, which restores the indifference on the price of the defaultable ZCB issued by I to the transition.

The proof can be generalized to multiple stochastic cash flows.

5: Bye Bye Multi-Curves ?

Past, present, and possible futures [1/3]



5: Bye Bye Multi-Curves ?

Past, present, and possible futures [2/3]

	Past	Present	Near Future	Far Future
Yield curves (single currency)	▪ EURIBOR	▪ OIS ▪ EURIBOR 1M ▪ EURIBOR 3M ▪ EURIBOR 6M ▪ EURIBOR 12M ▪ Basis	▪ ESTER ▪ OIS ▪ EURIBOR 1M ▪ EURIBOR 3M ▪ EURIBOR 6M ▪ EURIBOR 12M ▪ More basis	▪ ESTER
CCS basis yield curves	▪ Many CCS	▪ Many CCS	▪ Many CCS	▪ Many CCS
Cap/Floor Volatilities	▪ EURIBOR	▪ EURIBOR 1M ▪ EURIBOR 3M ▪ EURIBOR 6M ▪ EURIBOR 12M	▪ EURIBOR 1M ▪ EURIBOR 3M ▪ EURIBOR 6M ▪ EURIBOR 12M	▪ ESTER
Swaption volatilities	▪ EURIBOR	▪ EURIBOR 3M ▪ EURIBOR 6M	▪ EURIBOR 3M ▪ EURIBOR 6M	▪ ESTER

5: Bye Bye Multi-Curves ?

Past, present, and possible futures [3/3]



Once upon a time, there was an ancient financial world, where all the yield curves were the same, there were non bases, and just a couple of rate volatilities...

5: Bye Bye Multi-Curves ?



6: References

Regulations

1. H. Bartholomew (2019) "LCH plans 2020 switch to SOFR discounting", Risk.net, 12 February 2019.
2. CME Group (2018) "OTC SOFR Swaps Clearing", Q4 2018, <https://www.cmegroup.com/trading/interest-rates/cleared-otc-sofr-swaps.html>.
3. D. Duffie (2018) "Bloomberg Quantitative (BBQ) Seminar: How to build a strategy for converting LIBOR contracts", Bloomberg Professional Services, March 05, 2018
4. European Central Bank (2018) "Report by the working group on euro risk-free rates on the transition from EONIA to ESTER", Frankfurt, 20 December 2018.
5. European Central Bank (2019) "Feedback on the report on the transition from EONIA to ESTER by the working group on euro risk-free rates", Frankfurt, February 2019.
6. European Central Bank (2019) "Second public consultation by the working group on euro risk-free rates on determining an ESTER-based term structure methodology as a fallback in EURIBOR-linked contracts Summary of responses", Frankfurt, February 2019.
7. European Commission (2019) "Sustainable finance: Commission welcomes agreement on a new generation of low-carbon benchmarks", Press release, Brussels, 25 February 2019
8. Federal Reserve Bank of New York, <https://apps.newyorkfed.org/markets/autorates/fed%20funds>
9. ISDA (2018) "Benchmarks", November 2018.
10. J. Kes, (2018) "High level implementation plan", Working Group Euro Risk Free Rates, 18 October 2018.
11. LCH, (2018), "Consultation on transition to SOFR discounting", 6 November 2018, <https://www.lch.com/membership/ltd-membership/ltd-member-updates/consultation-transition-sofr-discounting>.
12. M. C. Lege (2018) "Nuova metodologia per la rilevazione dell'euribor e Risk Free Rates. Aggiornamento sulle evoluzioni in corso», Commissione Mercato Monetario", Assiom Forex, Milano 12 novembre 2018.
13. R. Mackenzie Smith (2019) "SOFR, so bad: liquidity lags transition ambitions", Risk.net, 4 February 2019.

6: References

14. P. Nicoloso (2018) "ESTER as an alternative euro risk-free rate", Presentation to ECB representative office, Brussels, 16 October 2018.
15. J. L. Schirmann (2018) "Update on the hybrid Euribor methodology", European Money Market Institute, Presentation to Working Group Euro Risk Free Rates, Frankfurt am Main, 18 October 2018.
16. Working Group on Euro Risk Free Rates Subgroup n. 2 (2018) "Update by Subgroup 2 on the identification and recommendation of a term structure on RFRs", 4th meeting, Frankfurt am Main, 11 July 2018.
17. Working Group on Euro Risk Free Rates Subgroup n. 2 (2018) "Update by Subgroup 2 on the identification and recommendation of a term structure on RFRs", 5th meeting, Frankfurt am Main, 13 September 2018.
18. Working Group on Euro Risk Free Rates Subgroup n. 2 (2018) "Update by Subgroup 2 on the identification and recommendation of a term structure on RFRs", 6th meeting, Frankfurt am Main, 18 October 2018.
19. Working Group on Euro Risk Free Rates Subgroup n. 4 (2018) "EONIA-RFR Transition", Presentation at the Roundtable on Euro Risk-Free Rates at the ECB, Frankfurt am Main, 9 November 2018.
20. Working Group on Euro Risk Free Rates Q&A Session (2018) "Discussion on term rates initiatives", Presentation at the Roundtable on Euro Risk-Free Rates at the ECB, Frankfurt am Main, 9 November 2018.

XVAs

21. Gregory, Jon, The XVA Challenge: Counterparty Credit Risk, Funding, Collateral, and Capital, John Wiley & Sons, September 2015
22. Green, Andrew, XVA: Credit, Funding and Capital Valuation Adjustment, John Wiley & Sons, November 2016
23. C. Albanese, S. Caenazzo, S. Iabichino (2015) "Capital and Funding", Working Paper.
24. L. Andersen, D. Duffie, Y. Song (2019) "Funding Value Adjustments", The Journal of Finance, Volume 74, Issue 1, February 2019.

7: Appendix

Funding spread [1]

- The value of the risky zero coupon under Recovery of treasury model

$$\mathcal{R}(t) = RP_d(t, T) \quad 0 \leq R \leq 1$$

becomes

$$\begin{aligned} P_I(t, T) &= R \mathbb{E}_t^{\mathbb{Q}} \left[\int_t^T P_d(u, T) \gamma(u) e^{\int_t^u (r(s) + \gamma(s)) ds} du + P_I(t, T) \right] \\ &= R \mathbb{E}_t^{\mathbb{Q}} \left[e^{\int_t^T r(s) ds} \int_t^T \gamma(s) e^{\int_t^u \gamma(s) ds} du \right] + P_I(t, T) \\ &= R \mathbb{E}_t^{\mathbb{Q}} \left[e^{\int_t^T r(s) ds} \left(1 - e^{\int_t^T \gamma(s) ds} \right) \right] + P_I(t, T) \\ &= R (P_d(t, T) - P_I(t, T)) + P_I(t, T) \\ &= R P_d(t, T) + (1 - R) + P_I(t, T) \\ &= R P_d(t, T) + (1 - R) P_d(t, T) S_I(t, T) \\ &= P_d(t, T) (R + (1 - R) S_I(t, T)) \end{aligned}$$

7: Appendix

Funding spread [2]

- We rewrite the ZCB value in such a way

$$\begin{aligned}P_I(t, T) &= P_d(t, T) [R + S_I(t, T) - RS_I(t, T) + 1 - 1] \\&= P_d(t, T) [1 - (1 - R) + S_I(t, T)(1 - R)] \\&= P_d(t, T) [1 - (1 - R)(1 - S_I(t, T))].\end{aligned}$$

- Since the zero funding spread is defined as

$$\mathcal{S}(t, T) = \frac{1}{T - t} \log \frac{P_d(t, T)}{P_I(t, T)}$$

then

$$\begin{aligned}\mathcal{S}(t, T) &= \frac{1}{T - t} \log \frac{1}{1 - (1 - R)(1 - S_I(t, T))} \\&= -\frac{1}{T - t} \log [1 - (1 - R)(1 - S_I(t, T))] \\&= -\frac{1}{T - t} \log [1 - 1 + S_I(t, T) + R - RS_I(t, T)] \\&= -\frac{1}{T - t} \log \left\{ S_I(t, T) \left[1 + R \left(-1 + \frac{1}{S_I(t, T)} \right) \right] \right\}.\end{aligned}$$

7: Appendix

Funding spread [3]

- We recall that the survival probability may be defined through the hazard rate $\gamma_I(t)$

$$S_I(t, T) = e^{-\int_t^T \gamma(s) ds}$$

- Inserting the survival probability inside the spread rate formula, we arrive to

$$\mathcal{S}(t, T) = \frac{\int_t^T \gamma(s) ds}{T - t} - \frac{1}{T - t} \log \left[1 + R \left(e^{\int_t^T \gamma(s) ds} - 1 \right) \right].$$

- In order to obtain the instantaneous spread rate, we apply the limit

$$s_I(t) = \lim_{T \rightarrow t^+} \mathcal{S}(t, T) = \gamma(t)(1 - R)$$